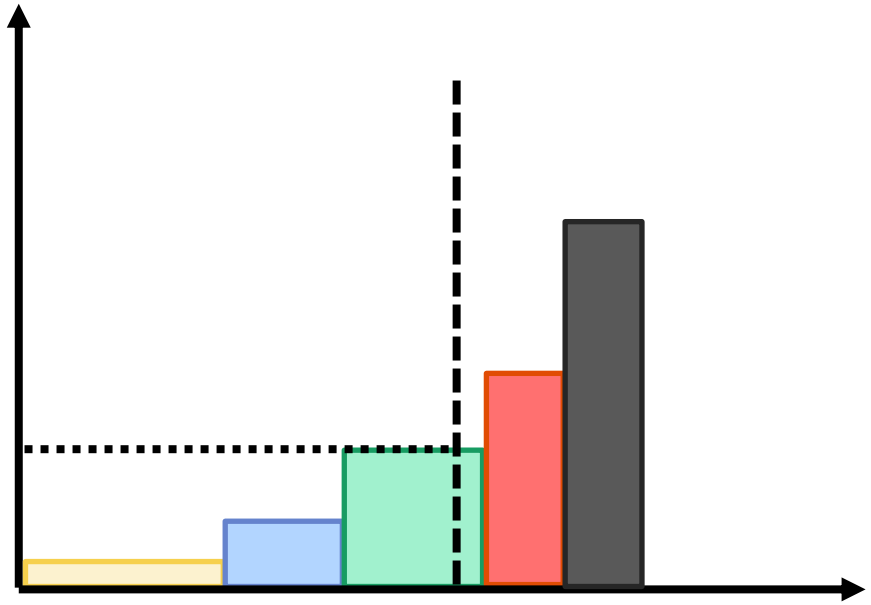
Game Theory in Electricity Markets

Week 3 – Solution Concepts 2

Nash equilibria in practice



	G1	G2	G3	G4	G5
Marginal cost	5	12.5	25	40	65
Capacity	100	40	50	30	30

Demand = 180

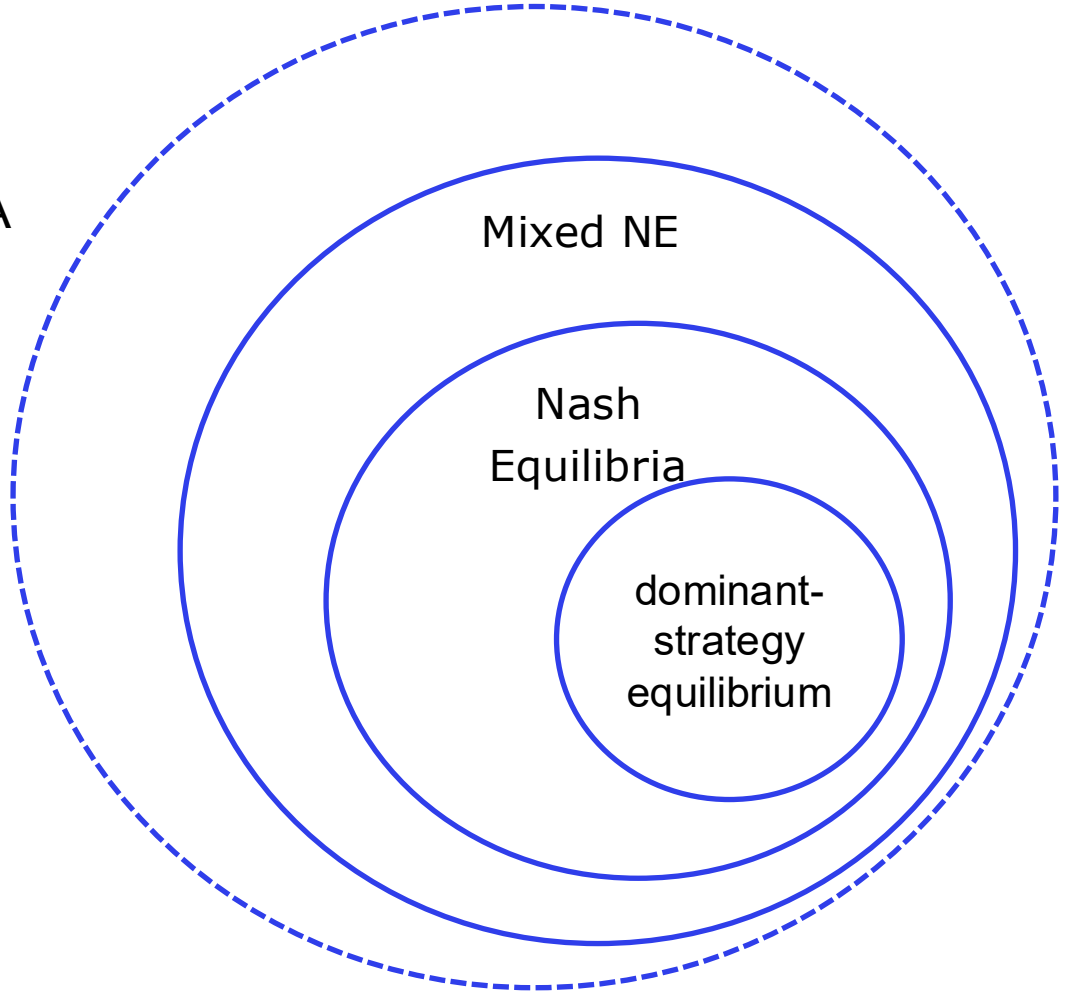
Integer bids in {0,…,80}

- There are many pure Nash equilibria
- We often see G1 or G2 or G3 = G5 -1 and rest very low. For example (1 ,2, 52, 3, 53)
- (80,0,0,0,0) is a Nash equilibrium

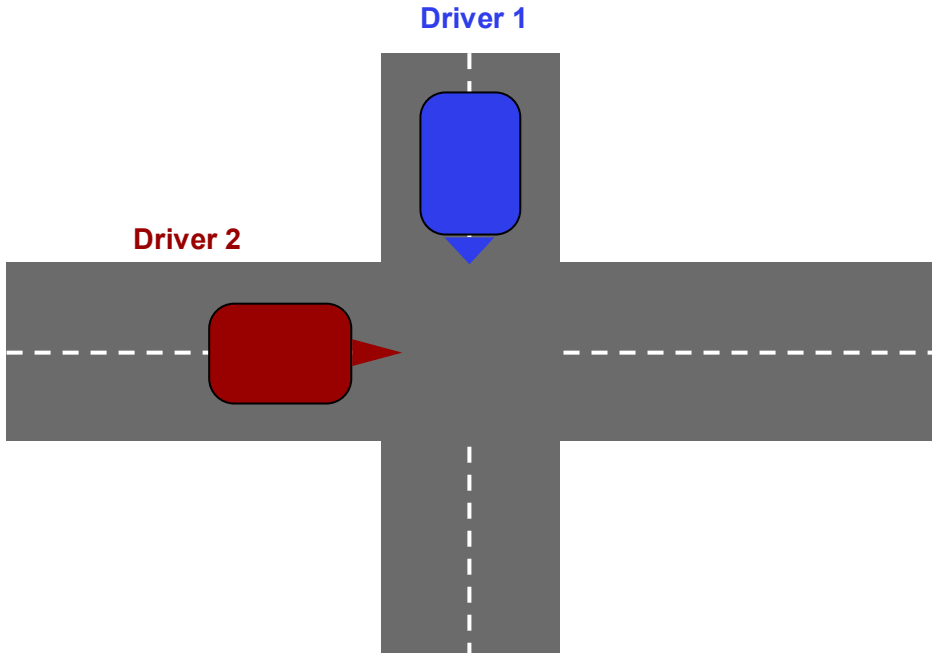
⟶ Weakly dominated actions can be part of a Nash equilibrium

Mixed Nash equilibrium, bad fact sheet

- To find it is PPAD hard (not in polynomial time)
  – How can players with simple algorithms even reach it?
- The set of mixed Nash equilibria is not arbitrary, but ugly
  – This is bad given that we want to optimize over it for PoA



Stop and Go



Two drivers approach the intersection at the same time.

Payoff matrix:

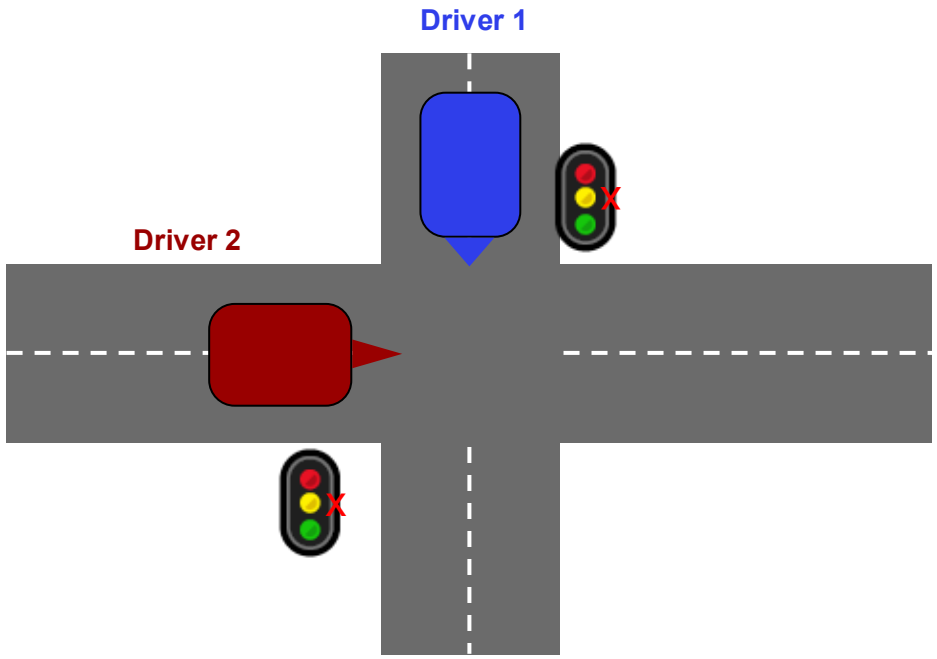
	Go	Stop
Go	-10, -10	2, 0
Stop	0, 2	0, 0

Mixed NE: 1/6 go, 5/6 stop

→ None of this is a good solution

Stop and Go

Idea: 🚦



- Introduce a trusted third party
- The trusted third party samples from a publicly know distribution
- The third part privately tells every player their suggestion
- Each player may follow this suggestion or not
- It is stable when the prior each player gets on the other players action leads to them picking the suggested action

Correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ over the joint action set A is a *correlated equilibrium* if no player can lower their expected cost by deviating from a privately recommended action:

$$E_{a\sim\pi}\,[c_i(a) \mid a_i] \;\leq\; E_{a\sim\pi}\,[c_i(a_i', a_{-i}) \mid a_i]$$

for every player $i \in N$ and every $a_i, a_i' \in A_i$.

A correlated equilibrium π that can be written as an independent (product) distribution, $\pi(a) = \sigma_1(a_1) \cdots \sigma_n(a_n)$, is a mixed Nash equilibrium.

The set of correlated equilibria

Proposition. In a finite game, the correlated-equilibrium condition is linear in π:

$$\Sigma_{a_{-i}} \pi(a_i, a_{-i})\, c_i(a_i, a_{-i}) \;\le\; \Sigma_{a_{-i}} \pi(a_i, a_{-i})\, c_i(a_i', a_{-i})$$

for every player $i \in N$ and every $a_i, a_i' \in A_i$.

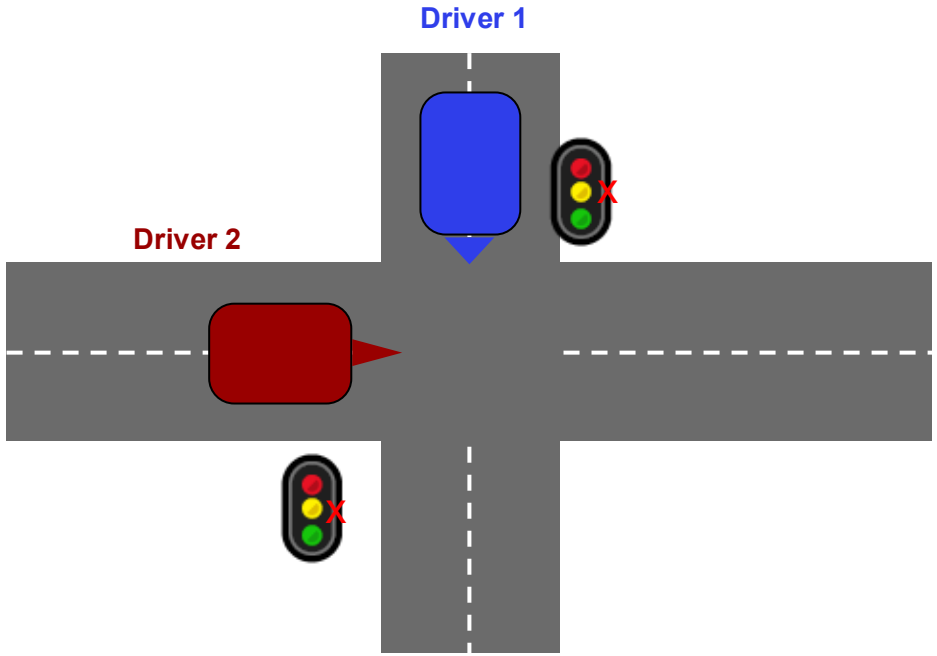
Together with $\pi(a) \ge 0$ and $\Sigma_{a \in A} \pi(a) = 1$, the correlated equilibrium set is:

$$\Pi_{CE}(\Gamma) = \{ \pi \in \Delta(A) : \Sigma_{a_{-i}} \pi(a_i, a_{-i}) [c_i(a_i, a_{-i}) - c_i(a_i', a_{-i})] \le 0 \quad \forall i, a_i, a_i' \}$$

Finitely many linear equalities and inequalities in $\pi \Rightarrow$ the correlated equilibrium set is a polytope.

The linearity argument stays true for infinite action spaces. It just has infinitely many inequalities within an infinite dimensional space.

Stop and Go



Two drivers approach the intersection at the same time.

Payoff matrix:

	Go	Stop
Go	-10, -10	2, 0
Stop	0, 2	0, 0

Do you see any Correlated equilibria?

The CE set for Stop and Go

Write $p_1=\pi(G,G)$, $p_2=\pi(G,S)$, $p_3=\pi(S,G)$, $p_4=\pi(S,S)$. Then $\Pi_{CE}(\Gamma)$ is given by:

IC (i=1, ai=G): $-10\,p_1 + 2\,p_2 \leq 0$

IC (i=1, ai=S): $10\,p_3 - 2\,p_4 \leq 0$

IC (i=2, ai=G): $-11\,p_1 + 2\,p_3 \leq 0$

IC (i=2, ai=S): $11\,p_2 - 2\,p_4 \leq 0$

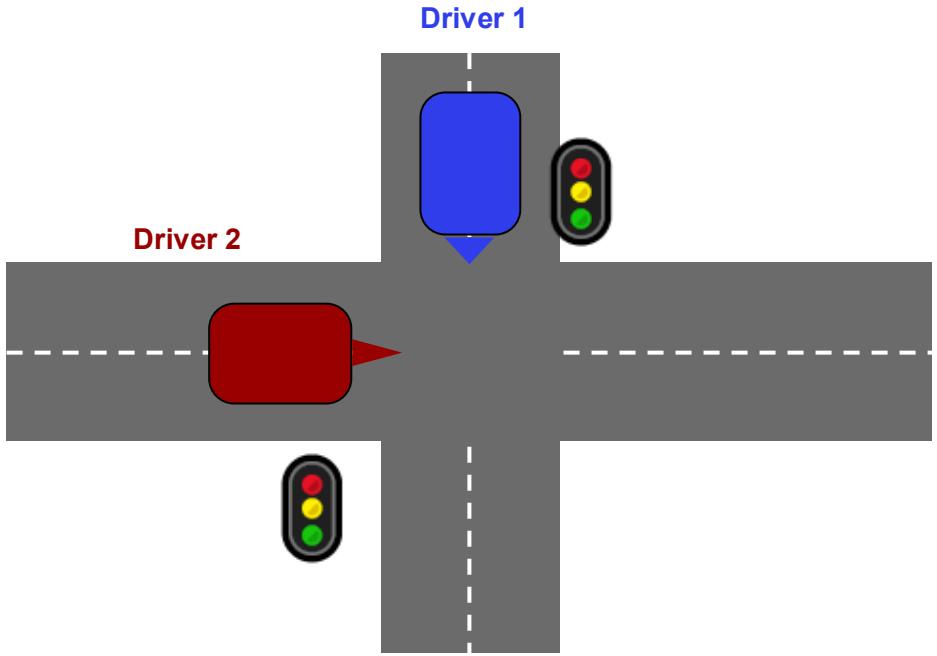
Normalization: $p_1 + p_2 + p_3 + p_4 = 1$

Nonnegativity: $p_1, p_2, p_3, p_4 \geq 0$

One inequality per pair of actions in $A_i \times A_i$.

Size grows O($|A_i|$*^2).*

Stop and Go and Slow



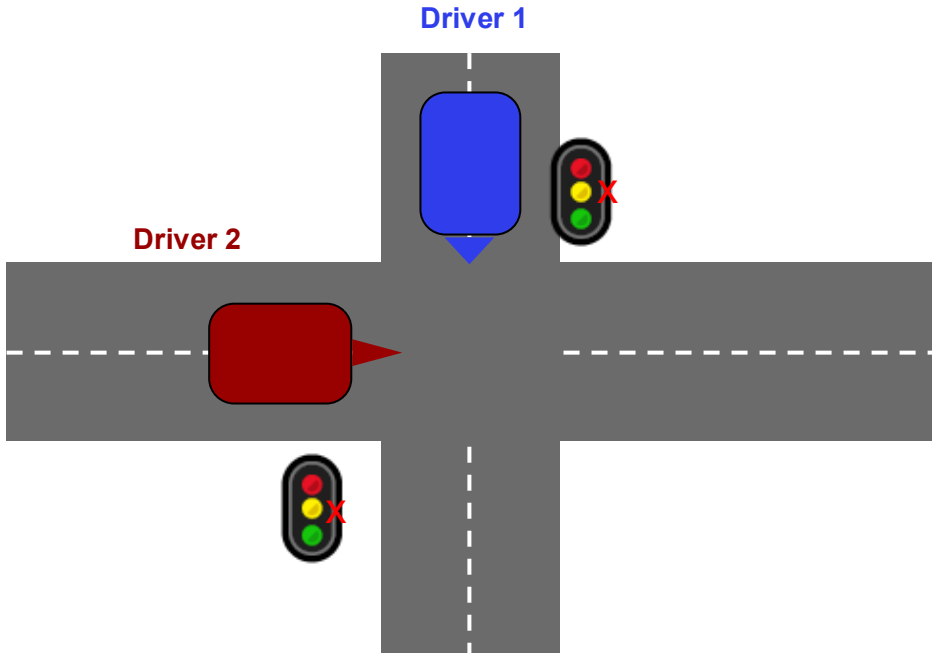
Two drivers approach the intersection at the same time.

Payoff matrix:

	Go	Slow	Stop
Go	-10, -10	-10, -10	2,0
Slow	-10, -10	-10, -10	1,0
Stop	0,2	0,1	0,0

Can a correlated equilibrium have full support?

Stop and Go and Slow



- Introduce a trusted third party
- The trusted third party publicly declares a distribution
- Each player needs to decide if they want to follow prior to drawing
- It is stable when the prior each player gets on the joint suggestion leads to them picking to follow

Coarse correlated equilibrium

Definition. A distribution $\pi \in \Delta(A)$ is a *coarse correlated equilibrium* if no player can lower their expected cost by an unconditional deviation to a fixed action:

$$E_{a\sim\pi}[c_i(a)] \;\le\; E_{a\sim\pi}[c_i(a_i', a_{-i})]$$

for every player $i \in N$ and every $a_i' \in A_i$.

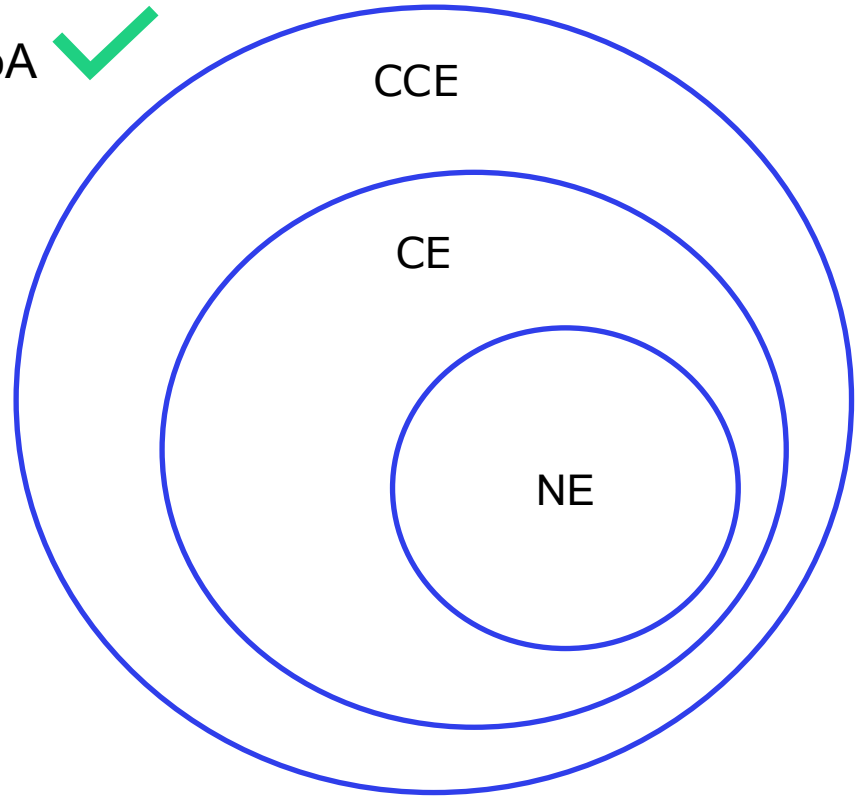
Both sides are already linear in π so the set

$$\Pi_{CCE}(\Gamma) = \{\, \pi \in \Delta(A) \;:\; \Sigma_{a\in A}\, \pi(a)\,[c_i(a) - c_i(a_i', a_{-i})] \le 0 \quad \forall\, i, a_i' \,\}$$

is again a polytope. Every correlated equilibrium is a coarse correlated equilibrium.

Mixed Nash equilibrium, bad fact sheet

- To find it is PPAD hard (not in polynomial time)
  - How can players with simple algorithms even reach it?
- The set of mixed Nash equilibria is not arbitrary, but ugly
  - This is bad given that we want to optimize over it for PoA ✓



Repeated game

Definition. The same stage game $\Gamma = (N, (A_i), (c_i))$ is played repeatedly, in rounds (time slots) $t = 1, \ldots, T$:

$a_i^t \in A_i$ player i's action in round t

Chosen simultaneously with the other players, possibly depending on the history of play up to round t.

$a^t = (a_1^t, \ldots, a_n^t) \in A$ the joint action in round t

Realized once every player has moved; player i incurs $c_i(a^t)$ and observes $c_i(a_i, a_{-i}^t)$ for every $a_i \in A_i$ (full information).

$\pi^T \in \Delta(A)$ the empirical distribution of play

$\pi^T(a) = (1/T) \Sigma_t 1[a^t = a]$

How to measure quality of a decision

Setup. A decision-maker repeatedly chooses among a set A of n = 2 actions, playing against an adversary. At each time t = 1, …, T:

$p_t \in \Delta(A)$ the decision-maker's strategy.

$c_t : A \to [0,1]$ the adversary's cost vector, chosen after seeing p_t

An action a_t is drawn according to p_t.

The decision-maker incurs cost $c_t(a_t)$ and learns the entire vector c_t, not just the realized cost (full information).

How do we evaluate the decision-maker? It is not fair to compare against the optimal in hindsight decision as the adversary could react to the decision-makers action.

No regret and no-regret algorithms

Definition. Player i's *regret* for not having played a fixed action $a_i' \in A_i$ throughout is:

$$Reg_i^T(a_i') = \Sigma_{t=1,\dots,T} [\, c_i(a_i^t, a_{-i}^t) - c_i(a_i', a_{-i}^t) \,]$$

the extra cost from following the realized play $a_i^1, \dots, a_i^T$ instead of a_i' throughout, holding the opponents' actual play fixed.

Player i has *no regret* if

$$\lim\sup_{T\to\infty} \; (1/T) \max_{ai' \in Ai} Reg_i^T(a_i') \leq 0$$

A no-regret algorithm maps histories to actions and guarantees this for every player, whatever the opponents do.

No-regret dynamics converge to CCE

Theorem. If every player follows a no-regret algorithm, every limit point of the empirical distribution π^T is a coarse correlated equilibrium of Γ.

Proof. Since π^T is the empirical distribution of (a_i^t, a_{-i}^t) over $t = 1, \ldots, T$:

$$\Sigma_{a \in A} \pi^T(a) [c_i(a) - c_i(a_i', a_{-i})] = (1/T) Reg_i^T(a_i')$$

for every $i \in N$ and $a_i' \in A_i$. Both sides are just the time-average of $c_i(a_i^t, a_{-i}^t) - c_i(a_i', a_{-i}^t)$ over the realized history.

Since player i has no regret, letting $T \to \infty$ gives:

$$\lim\sup_{T \to \infty} \Sigma_{a \in A} \pi^T(a) [c_i(a) - c_i(a_i', a_{-i})] \leq 0$$

for every $i \in N$ and a_i'. This is exactly the CCE condition. So every limit point π^* of (π^T) satisfies $\pi^* \in \Pi_{CCE}(\Gamma)$. ■

No-regret dynamics

- No-regret essentially means “I could not have done better playing one fixed action all the time”.
- This is a rather soft condition that most learning algorithms can assure.

- We can not hope to achieve much more as other players react to our strategy
- It can even be proven that generally no-regret algorithms need to accept some loss as one needs to randomize over actions

That each player achieves no-regret is neither to optimistic nor to pessimistic and therefore a realistic assumption

The multiplicative weights (MW) algorithm

Algorithm. Maintain a weight $w^t(a)$ for every action $a \in A$, initialized to $w^1(a) = 1$. At each time $t = 1, \ldots, T$:

$p^t := w^t/\Gamma^t$ play action a with probability $p^t(a)$

$\Gamma^t = \sum_{a \in A} w^t(a)$ is the sum of the weights => an action is played with probability proportional to its own weight.

$w^{t+1}(a) = w^t(a) \cdot (1 - \varepsilon)^{c^t(a)}$ update every action's weight using the cost vector c^t

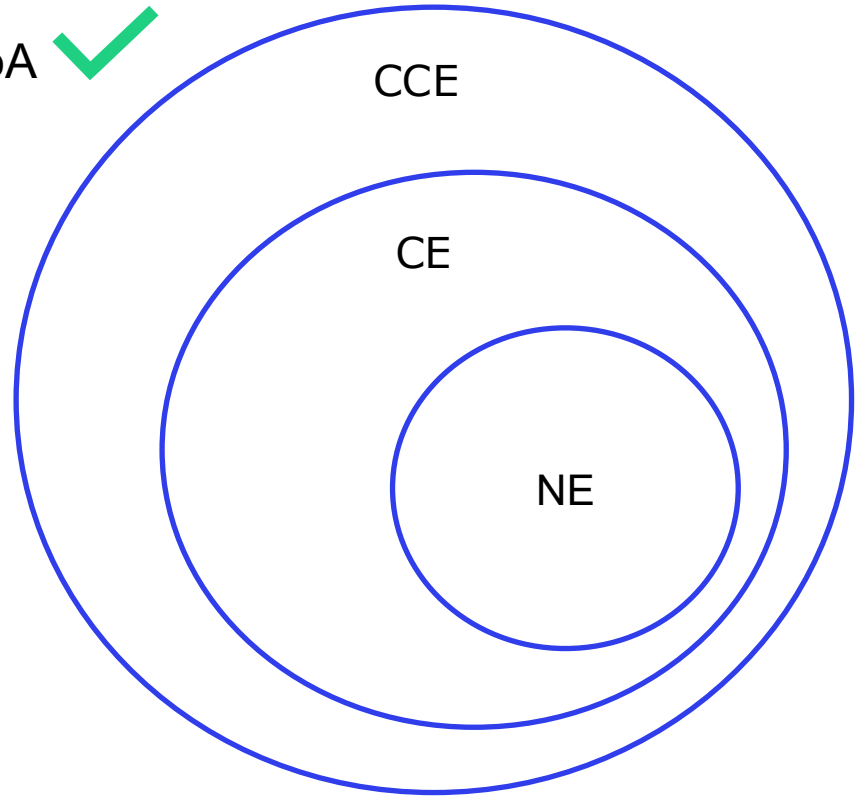
If costs are 0/1: a weight is unchanged when $c^t(a) = 0$, and multiplied by $(1 - \varepsilon)$ when $c^t(a) = 1$ — bad actions are punished at an exponential rate.

Role of $\varepsilon \in (0, \frac{1}{2}]$. Small $\varepsilon \to p^t$ close to uniform (exploration); $\varepsilon \to 1 \to$ all mass on the action with lowest cost so far (exploitation).

Guarantee. Setting $\varepsilon = \sqrt{(\ln n / T)}$ gives expected regret $\leq 2\sqrt{(\ln n / T)}$ with respect to every fixed action and only $O(\ln n / \varepsilon^2)$ rounds are needed for regret $\leq \varepsilon$. (This is provably as good as it gets.)

Mixed Nash equilibrium, bad fact sheet

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  - How can players with simple algorithms even reach it?
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Key take-aways

- Simple learning algorithms lead to (approximate) CCE's very quickly.
  - They are therefore surprisingly tractable and surprisingly realistic as predictions for game outcomes
- We moved away from considering the outcome of one game but or now considering the outcome of one game played multiple times.
  - We understand the learning trajectory of a game as its true outcome.
- What we do not know is if any CCE is reachable by learning dynamics and if so under how realistic ones.
  - This is the weakness in how realistic this equilibrium notation is.
- The set of all CCE or CE of a game can be described via linear inequalities
  - That's as nice as it gets, especially if one wants to optimize over them.


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